

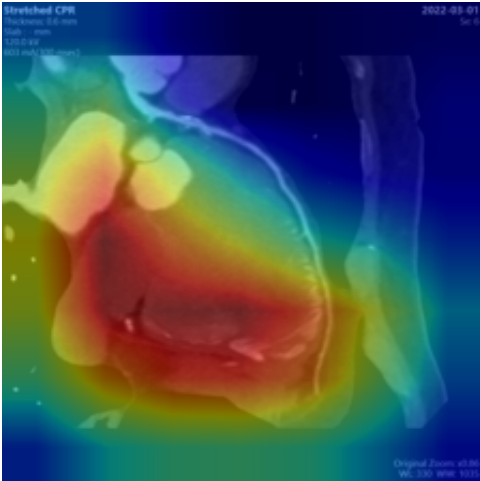
Clinical Diagnostic & RAG Report - Cad

Target Domain:	Cad	Image File:	abnormal k1_002.jpg
Predicted Class:	abnormal	Confidence:	99.4%
95% Conformal Set:	['normal', 'abnormal']	Uncertainty Status:	[WARNING] Human Review Recommended

Visual Diagnostic Imaging & Grad-CAM Heatmap



Original Medical Scan



Grad-CAM Saliency Overlay (ROI)

AI Clinical Explanation (FLAN-T5 Grounded RAG)

'The image was classified as 'abnormal' with 99.4% confidence. Biomedical literature (PMID:38408906) indicates: BACKGROUND: Coronary artery disease (CAD) is an important cause of global burden of disease. There is a paucity of data on the burden and risk factors for CAD in sub-Saharan Africa (SSA), despite the rising trends in the shared risk factors across regions. The recent introduction of cardiac catheterization laboratory services in SSA could shed light on the burden of CAD in the region.'

Faithfulness & Quantitative Text Metrics

NLI Faithfulness Score:	0.2489 / 1.0000	ROUGE-1:	0.8246
BLEU-4:	0.8243	BERTScore Similarity:	0.5539

Retrieved PubMed Literature Evidence

[1] Angiographic characteristics of coronary artery disease in patients undergoing diagnostic coronary angiography at a tertiary hospital in Tanzania. (PMID: 38408906)

'BACKGROUND: Coronary artery disease (CAD) is an important cause of global burden of disease. There is a paucity of data on the burden and risk factors for CAD in sub-Saharan Africa (SSA), despite the rising trends in the shared risk factors across regions. The recent introduction of cardiac catheterization laboratory services in SSA could shed light on the burden of CAD in the

region.'

[2] CAD-RADS™ 2.0 - 2022 Coronary Artery Disease - Reporting and Data System.: An expert consensus document of the Society of Cardiovascular Computed Tomography (SCCT), the American College of Cardiology (ACC), the American College of Radiology (ACR) and the North America Society of Cardiovascular Imaging (NASCI). (PMID: 36436841)

'Coronary Artery Disease Reporting and Data System (CAD-RADS) was created to standardize reporting system for patients undergoing coronary CT angiography (CCTA) and to guide possible next steps in patient management. The goal of this updated 2022 CAD-RADS 2.0 is to improve the initial reporting system for CCTA by considering new technical developments in Cardiac CT, including data from recent clinical trials and new clinical guidelines. The updated CAD-RADS classification will follow an established framework of stenosis, plaque burden, and modifiers, which will include assessment of lesion-specific ischemia using CT fractional-flow-reserve (CT-FFR) or myocardial CT perfusion (CTP), when performed.'

[3] CAD-RADS™ 2.0 - 2022 Coronary Artery Disease-Reporting and Data System: An Expert Consensus Document of the Society of Cardiovascular Computed Tomography (SCCT), the American College of Cardiology (ACC), the American College of Radiology (ACR), and the North America Society of Cardiovascular Imaging (NASCI). (PMID: 35864070)

'The main goal of CAD-RADS, which should always be interpreted together with the impression found in the report, remains to facilitate communication of test results with referring physicians along with suggestions for subsequent patient management. In addition, CAD-RADS will continue to provide a framework of standardization that may benefit education, research, peer-review, artificial intelligence development, clinical trial design, population health and quality assurance with the ultimate goal of improving patient care.'

[4] CAD-RADS™ 2.0 - 2022 Coronary Artery Disease-Reporting and Data System: An Expert Consensus Document of the Society of Cardiovascular Computed Tomography (SCCT), the American College of Cardiology (ACC), the American College of Radiology (ACR), and the North America Society of Cardiovascular Imaging (NASCI). (PMID: 36115815)

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